Capstone — Low-rank frames vs differences

Launch Fri Dec 4 · Studio Mon Dec 7 · Gallery Wed Dec 9

Last updated: October 8, 2026.

Dates are on the schedule. There is **no quiz on SVD**. Friday Dec 4 is **project launch** (this packet) with the SVD lecture and Lab 6 (Fit and Compress). The graded product is this project (Canvas file upload).

Octave: Lachniet Ch. 5 (SVD) and Ch. 7 (image compression). Python: NumPy `svd` plus a simple image/array load. Syntax pointers: Resources.

The question

You get (or build) **three** grayscale frames F_1, F_2, F_3, each an $m \times n$ matrix of intensities.

Compare two encoding strategies with rank-k SVD approximations:

Plan	What you compress with a rank-k SVD
A — Independent frames	F_1, F_2, and F_3 separately
B — Frame + differences	F_1, then $D_2 = F_2 - F_1$, then $D_3 = F_3 - F_2$

Driving question: For your data, when (if ever) is Plan B better than Plan A on **accuracy**, **storage**, and **runtime / resource use** — including the cost of computing the SVDs?

Computing an SVD is **not** free. Low-rank storage or faster later multiplies only help when the matrix is well approximated at modest k **and** you reuse the representation enough (or care enough about storage/transmission) to justify the factorization cost. Image examples are instructional.

Timeline

When	What
Fri Dec 4	**Project launch** in Lecture 33 (after Quiz 11 / with Lab 6). Skim this packet; start choosing data.
Mon Dec 7	Studio — implement, measure, debug.
Wed Dec 9	Gallery — short presentation or station; **Canvas upload due** (end of day unless announced otherwise).

What to turn in (Canvas)

One PDF (or short report + clearly labeled figures), **individual** submission, about **1–2 pages of writing** plus figures/tables:

1. **Data.** Where the three frames came from (phone stills, course sample, synthetic frames, etc.) and size $m \times n$.
2. **Methods.** Rank(s) k you tried; how you formed D_2, D_3; how you rebuilt approximate frames under Plan B.
3. **Evidence (required numbers).**
   - Relative reconstruction error vs k for Plan A and Plan B (plot or table).
   - Storage count: full mn vs rank-k cost $\approx k(m + n + 1)$ per compressed matrix (and totals for A vs B).
   - Timing: at least one honest runtime comparison (e.g. SVD time vs applying a low-rank map, or Plan A vs B factorization time for the same k).
4. **Conclusion.** One clear recommendation: for *your* frames, prefer A, prefer B, or "depends," with a quantitative reason and **one limitation**.

Gallery: 2–3 minute lightning talk **or** a station with one figure people can read in 30 seconds. Peers may ask about your numbers.